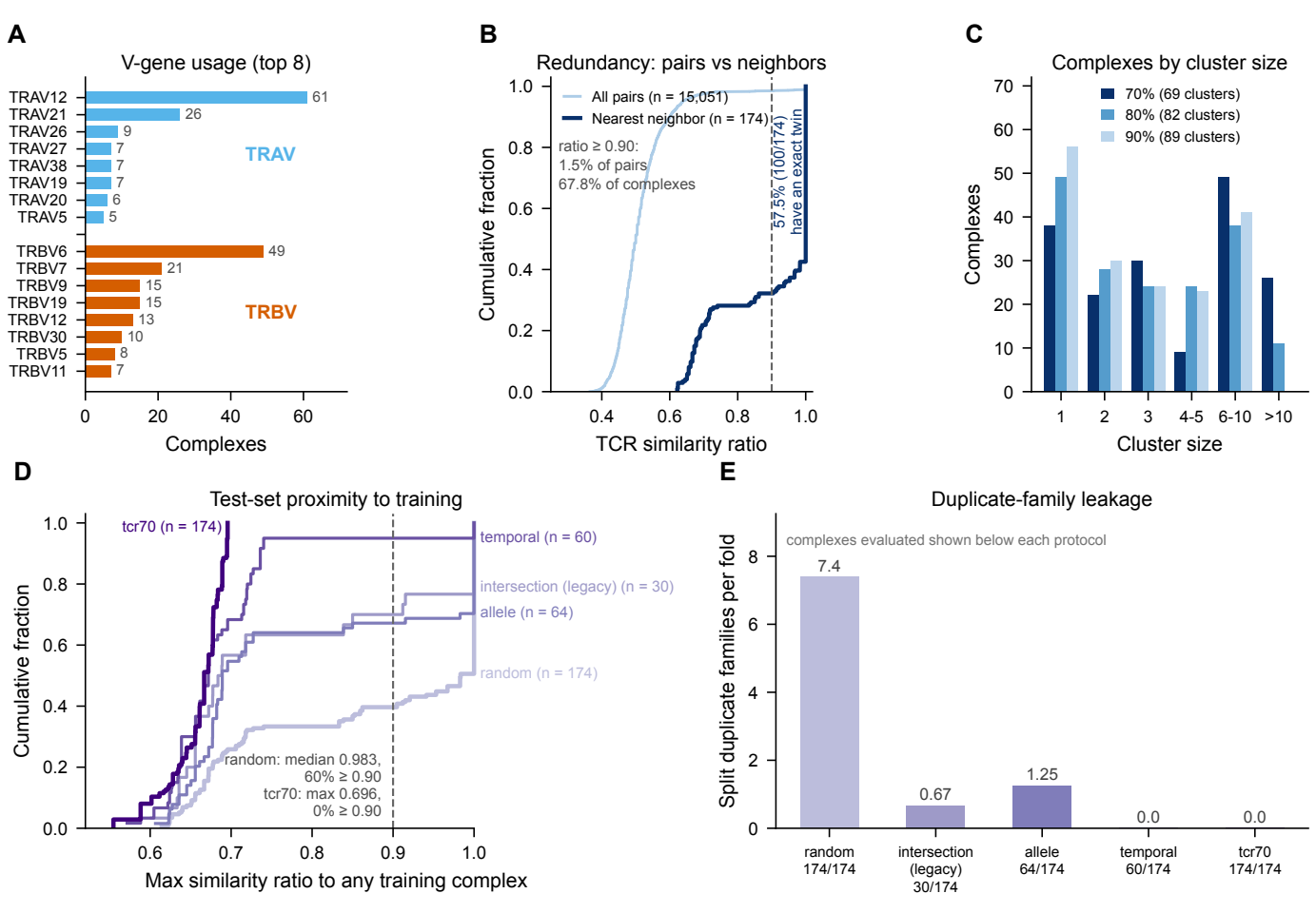
A
V-gene usage (top 8)
TRAV12
TRAV21
TRAV26
TRAV27
TRAV38
TRAV19
TRAV20
TRAV5
61
26
9
7
7
7
6
5
TRAV
TRBV6
TRBV7
TRBV9
TRBV19
TRBV12
TRBV30
TRBV5
TRBV11
49
21
15
15
13
10
8
7
TRBV
0
20
40
60
Complexes
B
Redundancy: pairs vs neighbors
All pairs (n = 15,051)
Nearest neighbor (n = 174)
ratio ≥ 0.90:
1.5% of pairs
67.8% of complexes
57.5% (100/174)
have an exact twin
Cumulative fraction
0.0
0.2
0.4
0.6
0.8
1.0
0.4
0.6
0.8
1.0
TCR similarity ratio
C
Complexes by cluster size
70% (69 clusters)
80% (82 clusters)
90% (89 clusters)
Complexes
0
10
20
30
40
50
60
70
1
2
3
4-5
6-10
>10
Cluster size
D
Test-set proximity to training
tcr70 (n = 174)
temporal (n = 60)
intersection (legacy) (n = 30)
allele (n = 64)
random (n = 174)
random: median 0.983,
60% ≥ 0.90
tcr70: max 0.696,
0% ≥ 0.90
Cumulative fraction
0.0
0.2
0.4
0.6
0.8
1.0
0.6
0.7
0.8
0.9
1.0
Max similarity ratio to any training complex
E
Duplicate-family leakage
complexes evaluated shown below each protocol
Split duplicate families per fold
0
2
4
6
8
7.4
0.67
1.25
0.0
0.0
random
174/174
intersection
(legacy)
30/174
allele
64/174
temporal
60/174
tcr70
174/174